

Curriculum Outline

Web Development Fundamentals · AfricaPlan Foundation
3 Courses · 12 Modules · 60 Lessons · 44 Hours Total

Program Overview

The HackathonAfrica curriculum is a beginner-friendly, three-course web development track designed to take students from zero coding experience to building real websites. Upon successful completion, students become eligible for HackathonAfrica hackathon opportunities.

Eligibility Requirements

- Complete all 3 courses
- Pass all module quizzes (minimum score: 70%)
- Pass all final exams (minimum score: 70%)
- Receive admin approval (optional additional review)

Course 1: HTML — Mastering the Web's Backbone

12 hours | Beginner | 20 Lessons | 4 Modules

Learn the building blocks of the web. HTML gives structure to every page you visit online. Students go from zero to confidently writing HTML documents with practical exercises and real-world examples.

Module 1: Document Structure & Core Elements

Learn the anatomy of HTML documents, essential tags, and how browsers interpret your code.

#	Lesson	Duration
1	How the Web Works	15 min
2	Setting Up Your Development Environment	15 min
3	HTML Document Structure	15 min
4	Headings, Paragraphs & Text	15 min
5	Links and Navigation	15 min

Assessment: Document Structure Quiz — 5 questions, 70% pass mark

Module 2: Forms & User Input

Master HTML forms, input types, validation, and creating interactive user experiences.

#	Lesson	Duration
1	Introduction to HTML Forms	15 min
2	Input Types	15 min
3	Form Validation	15 min
4	Form Accessibility & UX	15 min
5	Advanced Form Features	15 min

Assessment: Forms & Input Quiz — 5 questions, 70% pass mark

Module 3: Semantic HTML & Accessibility

Write meaningful HTML that search engines love and screen readers understand.

#	Lesson	Duration
1	What is Semantic HTML?	15 min
2	Page Structure Elements	15 min
3	Lists and Tables	15 min
4	Images and Media	15 min
5	ARIA and Accessibility	15 min

Assessment: Semantic HTML Quiz — 5 questions, 70% pass mark

Module 4: Advanced HTML & Performance

HTML5 APIs, media elements, performance optimization, and modern HTML features.

#	Lesson	Duration
1	HTML5 Media Elements	15 min
2	Canvas and SVG	15 min
3	Meta Tags and SEO	15 min
4	Web Components Basics	15 min
5	Performance Optimization	15 min

Assessment: Advanced HTML Quiz — 5 questions, 70% pass mark

Final Exam: HTML Final Assessment — 45 min | 15 MCQ + 2 Coding Exercises | 70% pass mark

Course 2: CSS — Styling the Modern Web

14 hours | Beginner | 20 Lessons | 4 Modules

Transform plain HTML into beautiful, professional websites. CSS controls colours, layouts, fonts, spacing, and animations. This course takes students from basic styling to modern responsive design.

Module 1: Selectors, Box Model & Typography

Master CSS selectors, understand the box model, and create beautiful typography.

#	Lesson	Duration
1	CSS Selectors	15 min
2	The Box Model	15 min
3	Typography	15 min
4	Colors and Backgrounds	15 min
5	Pseudo-classes and Pseudo-elements	15 min

Assessment: Selectors & Box Model Quiz — 5 questions, 70% pass mark

Module 2: Layout — Flexbox & Grid

Modern CSS layout techniques for responsive, flexible designs.

#	Lesson	Duration
1	Flexbox Fundamentals	15 min
2	Flex Item Properties	15 min
3	CSS Grid Fundamentals	15 min
4	Grid Template Areas	15 min
5	Responsive Design	15 min

Assessment: Layout Quiz — 5 questions, 70% pass mark

Module 3: Animations, Transitions & Effects

Bring your designs to life with CSS animations and visual effects.

#	Lesson	Duration
1	CSS Transitions	15 min
2	CSS Animations	15 min
3	Transforms	15 min
4	Shadows and Effects	15 min
5	Filters and Blend Modes	15 min

Assessment: Animations Quiz — 5 questions, 70% pass mark

Module 4: Advanced CSS Architecture

CSS variables, methodologies, and scalable architecture patterns.

#	Lesson	Duration
1	CSS Variables (Custom Properties)	15 min
2	CSS Methodologies (BEM)	15 min
3	CSS Architecture Patterns	15 min
4	Modern CSS Features	15 min
5	CSS Best Practices	15 min

Assessment: CSS Architecture Quiz — 5 questions, 70% pass mark

Final Exam: CSS Final Assessment — 45 min | 15 MCQ + 2 Coding Exercises | 70% pass mark

Course 3: JavaScript — From Fundamentals to Mastery

18 hours | Beginner | 20 Lessons | 4 Modules

Bring your websites to life with interactivity. JavaScript is the programming language of the web — it handles user input, animations, data processing, and much more.

Module 1: JavaScript Fundamentals

Variables, data types, operators, control flow, and functions.

#	Lesson	Duration
1	Variables and Data Types	15 min
2	Operators and Expressions	15 min
3	Control Flow	15 min
4	Loops	15 min
5	Functions	15 min

Assessment: JS Fundamentals Quiz — 5 questions, 70% pass mark

Module 2: DOM & Browser APIs

Manipulate web pages, handle events, and work with browser features.

#	Lesson	Duration
1	DOM Selection	15 min
2	DOM Manipulation	15 min
3	Event Handling	15 min
4	Creating Elements	15 min
5	Forms and Validation	15 min

Assessment: DOM Quiz — 5 questions, 70% pass mark

Module 3: Asynchronous JavaScript

Callbacks, promises, async/await, and fetching data.

#	Lesson	Duration
1	Callbacks	15 min
2	Promises	15 min
3	Async/Await	15 min
4	Fetch API	15 min
5	Error Handling	15 min

Assessment: Async Quiz — 5 questions, 70% pass mark

Module 4: Modern JavaScript & Patterns

ES6+ features, modules, and common design patterns.

#	Lesson	Duration
1	Destructuring	15 min
2	Spread and Rest	15 min
3	Modules	15 min
4	Classes	15 min
5	Best Practices	15 min

Assessment: Modern JS Quiz — 5 questions, 70% pass mark

Final Exam: JavaScript Final Assessment — 60 min | 15 MCQ + 3 Coding Exercises | 70% pass mark

Curriculum Summary

Course	Modules	Lessons	Duration	Final Exam
HTML — Mastering the Web's Backbone	4	20	12 hrs	45 min
CSS — Styling the Modern Web	4	20	14 hrs	45 min
JavaScript — From Fundamentals to Mastery	4	20	18 hrs	60 min
TOTAL	12	60	44 hrs	—

Assessment Structure

- Module Quizzes: 12 quizzes (one per module), 5 questions each, 70% minimum pass mark
- Final Exams: 3 exams (one per course), MCQ + coding exercises, 70% minimum pass mark
- Interactive Exercises: Hands-on coding exercises embedded throughout each lesson
- Live Preview: Students see their code rendered in real time

Learning Outcomes

By completing this curriculum, students will be able to:

- Build structured, semantic web pages using HTML5

- Style and layout pages with modern CSS including Flexbox and Grid
- Add interactivity with JavaScript including DOM manipulation and async data fetching
- Apply web accessibility and performance best practices
- Work confidently in a real development environment

*HackathonAfrica is a program of AfricaPlan Foundation — africaplanfoundation.org
Empowering African youth through technology education and hackathon opportunities.*